

Face Occlusion Prediction

Model Training Track — Data Challenge 2026

OccluVision (Group 8)

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<https://github.com/LeHoang510/Face-Occlusion-Prediction>

Introduction

Face occlusion estimation quantifies how much of a detected face is hidden (masks, hands, objects), which matters for downstream biometrics and fairness-aware systems. In the Model Training Track we regress a continuous occlusion ratio in $[0, 1]$ from 224×224 RGB face crops. The evaluation metric penalises errors on heavily occluded samples and rewards balanced performance across genders, so accuracy alone is insufficient. We pursued a three-phase pipeline: (i) benchmark diverse CNN/ViT backbones, (ii) test stronger training and architectural variants including balanced batching, frozen backbones, segmentation-aware features, and multitask heads, and (iii) fine-tune a large self-supervised encoder with parameter-efficient adapters. Our submitted model—DINOv3-L + LoRA—ranks **3rd** on the interim leaderboard (score **0.00099**); code is available at the repository link above.

Task & Metric

The leaderboard score combines weighted accuracy and gender fairness (lower is better):

$$w_i = \frac{1}{30} + \text{GT}_i, \quad \text{Err} = \frac{\sum_i w_i (p_i - \text{GT}_i)^2}{\sum_i w_i}, \quad \text{Score} = \frac{\text{Err}_F + \text{Err}_M}{2} + |\text{Err}_F - \text{Err}_M|.$$

Weights penalise errors on heavily occluded faces; the disparity term encourages balanced performance across genders. All models optimise the same weighted MSE objective with standard augmentations (resize, flip, colour jitter). During development we hold out 10% of labelled data for validation (seed 42); the submitted checkpoint is retrained on the *full* training set.

Methods Explored

Phase 1 — CNN/ViT baselines. We fine-tune ImageNet-pretrained backbones from `timm/torchvision`: ResNet-50, ConvNeXt-Tiny, Swin-T, ViT-B/16, EfficientViT-B1, and DINOv2-Small. Each replaces the classification head with a dropout-regularised MLP ending in Sigmoid; all backbone weights are updated (up to 20 epochs, batch size 32, AdamW with cosine decay).

Phase 2 — Advanced experiments. Beyond plain backbone fine-tuning, we tested four extensions: balanced-gender mini-batches to reduce the $|\text{Err}_F - \text{Err}_M|$ penalty, frozen-backbone variants with lightweight heads, segmentation-aware auxiliary features from person masks, and multitask heads that jointly predict occlusion and gender. Table 1 reports the preserved validation results.

Phase 3 — DINOv3-L + LoRA. We adopt `facebook/dinov3-vitl16-pretrain-lvd1689m` (ViT-L/16), freeze the backbone, and inject LoRA adapters ($r=16$) on attention projections. Only LoRA weights and a regression head are trained—a parameter-efficient alternative to full fine-tuning. We train both a standard random sampler and a balanced-gender variant; Table 1 reports validation scores from the development split. The final submission retraining the best DINOv3-L + LoRA configuration on all labelled training data.

Submitted Solution: DINOv3-L + LoRA

Architecture. We freeze `facebook/dinov3-vitl16-pretrain-lvd1689m` (ViT-L/16, 1.1B params) and inject LoRA adapters ($r=16$, $\alpha=32$, dropout 0.05) on `q_proj` and `v_proj`. The CLS token feeds a regression head: LayerNorm $\rightarrow$ MLP(512, GELU) $\rightarrow$ Linear $\rightarrow$ Sigmoid. Only $\sim 2\%$ of parameters are trainable (LoRA + head).

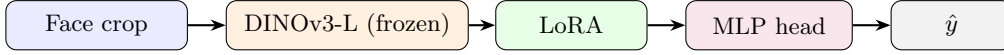


Figure 1: Submitted pipeline: frozen DINOv3-L backbone with LoRA adapters and a lightweight MLP head.

Training. We first tune hyper-parameters with a 90/10 train/val split, then retrain on the full labelled training set for submission. Training uses AdamW ($\text{lr}=5\times 10^{-4}$, weight decay 10^{-4}), cosine schedule with 1-epoch warmup, gradient clipping at 1.0; 15 epochs, batch size 64 on one RTX 3090 (~ 6 h). The development run reaches its best validation score at epoch 9 (Table 1); this configuration is reused for the full-data retraining that produces our leaderboard submission.

Results

Table 1: Best preserved validation scores during model selection (90/10 split; lower is better).

Model	Score	Err _F	Err _M	Ep.
<i>Phase 1 — CNN/ViT baselines</i>				
Swin-T	0.001090	0.000481	0.000887	15
ConvNeXt-Tiny	0.001112	0.000671	0.000965	4
ViT-B/16	0.001162	0.000563	0.000962	9
ResNet-50	0.001221	0.000536	0.000992	12
EfficientViT-B1	0.001285	0.000660	0.001076	9
DINOv2-Small	0.001469	0.000693	0.001210	19
<i>Phase 2 — advanced experiments</i>				
ConvNeXt-Tiny (balanced)	0.001097	0.000462	0.000885	14
ResNet-50 (balanced)	0.001134	0.000492	0.000920	14
Swin-T (balanced)	0.001150	0.000547	0.000949	11
ViT-B/16 (balanced)	0.001157	0.000628	0.000980	7
ResNet-50 (multitask)	0.001318	0.000659	0.001103	17
Swin-T (segmentation-aux)	0.002672	0.002660	0.002672	20
ConvNeXt-Tiny (segmentation-aux)	0.002777	0.002139	0.002565	20
<i>Phase 3 — DINOv3 (submitted model in bold)</i>				
DINOv3-L + LoRA	0.001059	0.000567	0.000895	9
DINOv3-L + LoRA (balanced)	0.001129	0.000525	0.000928	15
DINOv3-L (full fine-tuning)	0.001216	0.000693	0.001041	9

Limitations & Outlook

Limitations.

- **Compute:** Limited training budget—all experiments were capped at 20 epochs, which may leave performance on the table.
- **Hyper-parameters:** LoRA rank, learning rate, batch size, and epoch count all affect the fairness–accuracy trade-off; we did not run a full grid search.
- **Gender gap:** Despite the metric penalty, male error remains higher than female error on validation.
- **Incomplete archives:** Some frozen-backbone, segmentation, and multitask runs were configured but their full metrics were not preserved locally, so Table 1 includes only recoverable results.

Future work.

- **Close the gender gap:** combine LoRA fine-tuning with balanced-gender batching (which lowered Err_F in isolation) and tune the loss to better match the $|\text{Err}_F - \text{Err}_M|$ penalty.
- **Ensemble:** average or stack predictions from DINOv3-L + LoRA with strong CNN/ViT baselines (e.g. Swin-T) to exploit complementary errors.